POLYP DETECTION MEDICAL REPORT

Analysis Date: June 04, 2026

VIDEO INFORMATION

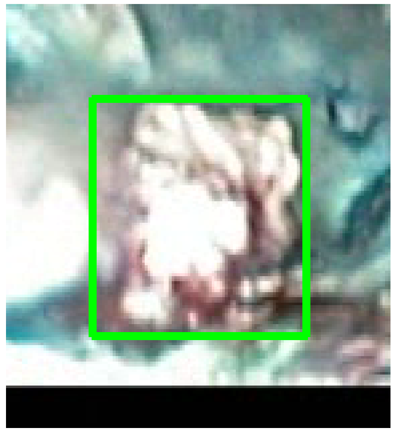
Video ID:	a6337f30-3932-4dc1-9563-5f93872cfef8
Total Frames:	2974
Frame Rate:	25.0 fps (assumed)
Duration:	119.0 seconds

DETECTION SUMMARY

Detection Method	Count	Status
YOLO Detections	198	✓
RT-DETR Detections	1860	✓
MedSAM2 Detections	1831	✓
Consensus (3-Model Agreement)	15	✓
Risk Classification	**Count**	**Percentage**
HIGH RISK	0	0.0%
MEDIUM RISK	0	0.0%
LOW RISK	0	0.0%
TOTAL ANALYZED	15	100.0%

DETAILED POLYP ANALYSIS

POLYP #1 — LIFTED_POLYP [LOW_RISK] (Tracked across 22 frames)

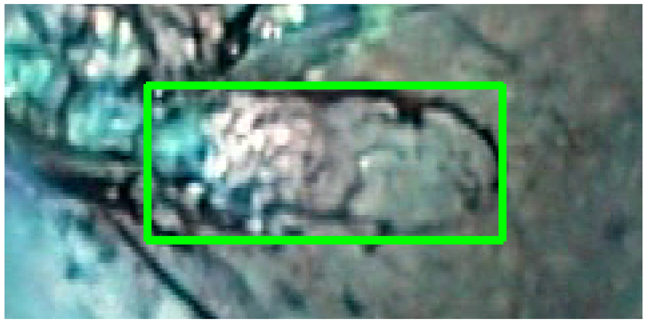


Feature	Value	Interpretation
Frame Index	287	Frame 287 of 2974
Detection Method	Detector Consensus (YOLO + RT-DETR agreement (MedSAM2 unavailable for this video)	(MedSAM2 Unavailable)
POLYP TYPE	**LIFTED_POLYP**	Type confidence: 72.7%
Redness Score	0.000	Color-based vascularization
Relative Radius	6.5%	Polyp size as % of frame diagonal
Texture Score	0.460	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	67.4%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #2 — LIFTED_POLYP [LOW_RISK] (Tracked across 203 frames)

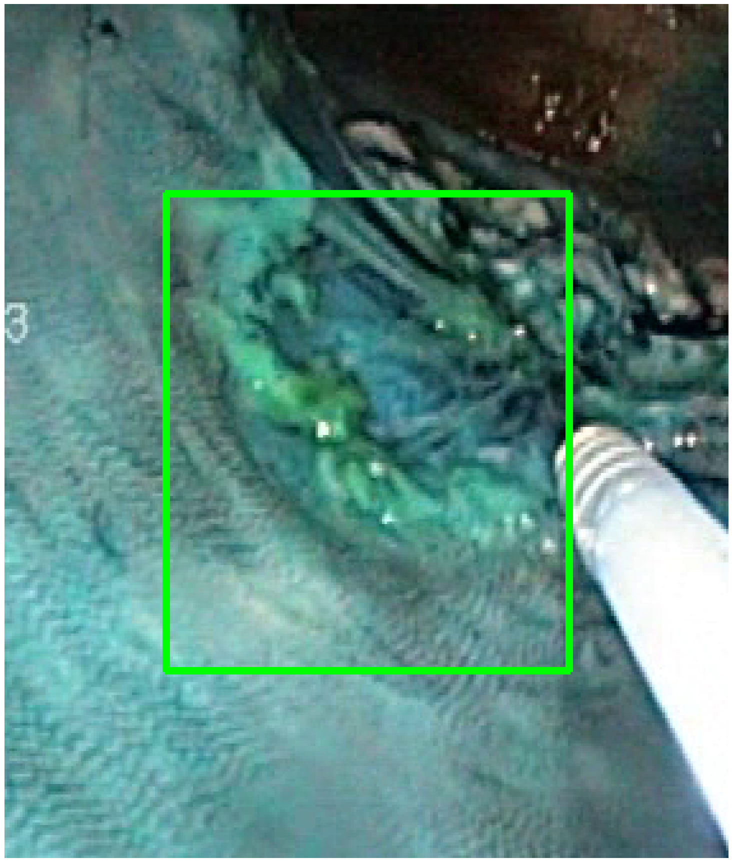


Feature	Value	Interpretation
Frame Index	2513	Frame 2513 of 2974
Detection Method	Detector Consensus (YOLO + RT-DETR agreement (MedSAM2 unavailable for this video)	(MedSAM2 Unavailable)
POLYP TYPE	**LIFTED_POLYP**	Type confidence: 74.6%
Redness Score	0.002	Color-based vascularization
Relative Radius	15.1%	Polyp size as % of frame diagonal
Texture Score	0.517	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	69.4%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #3 — LIFTED_POLYP [LOW_RISK] (Tracked across 154 frames)



Feature	Value	Interpretation
Frame Index	526	Frame 526 of 2974
Detection Method	Detector Consensus (YOLO + RT-DETR agreement (MedSAM2 unavailable for this video)	(MedSAM2 Unavailable)
POLYP TYPE	**LIFTED_POLYP**	Type confidence: 78.8%
Redness Score	0.000	Color-based vascularization
Relative Radius	15.5%	Polyp size as % of frame diagonal
Texture Score	0.429	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified

Detection Confidence	61.0%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #4 — LIFTED_POLYP [LOW_RISK] (Tracked across 20 frames)

Feature	Value	Interpretation
Frame Index	982	Frame 982 of 2974
Detection Method	Detector Consensus (YOLO + RT-DETR agreement)	(MedSAM2 unavailable for this video)
POLYP TYPE	**LIFTED_POLYP**	Type confidence: 81.3%
Redness Score	0.000	Color-based vascularization
Relative Radius	5.2%	Polyp size as % of frame diagonal
Texture Score	0.472	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	55.9%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

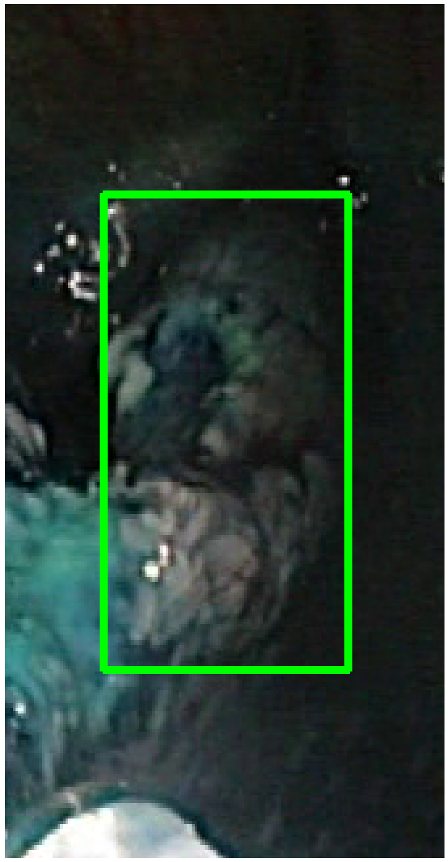
POLYP #5 — LIFTED_POLYP [LOW_RISK] (Tracked across 38 frames)

Feature	Value	Interpretation
Frame Index	1450	Frame 1450 of 2974
Detection Method	Detector Consensus (YOLO + RT-DETR agreement (MedSAM2 unavailable for this video)	MedSAM2 Unavailable
POLYP TYPE	**LIFTED_POLYP**	Type confidence: 85.2%
Redness Score	0.000	Color-based vascularization
Relative Radius	8.9%	Polyp size as % of frame diagonal
Texture Score	0.589	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	57.5%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #6 — LIFTED_POLYP [LOW_RISK] (Tracked across 25 frames)



Feature	Value	Interpretation
Frame Index	1598	Frame 1598 of 2974
Detection Method	Detector Consensus (YOLO + RT-DETR agreement (MedSAM2 unavailable for this video)	MedSAM2 Unavailable
POLYP TYPE	**LIFTED_POLYP**	Type confidence: 64.3%
Redness Score	0.001	Color-based vascularization
Relative Radius	14.5%	Polyp size as % of frame diagonal
Texture Score	0.204	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	LOW_RISK	Final symbolic reasoning bucket
Expert Prediction	LOW_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified

Detection Confidence	60.9%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #7 — LIFTED_POLYP [LOW_RISK] (Tracked across 24 frames)

Feature	Value	Interpretation
Frame Index	2597	Frame 2597 of 2974
Detection Method	Detector Consensus (YOLO + RT-DETR agreement) (MedSAM2 unavailable for this video)	MedSAM2 Unavailable
POLYP TYPE	**LIFTED_POLYP**	Type confidence: 76.3%
Redness Score	0.003	Color-based vascularization
Relative Radius	15.4%	Polyp size as % of frame diagonal
Texture Score	0.577	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	73.6%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

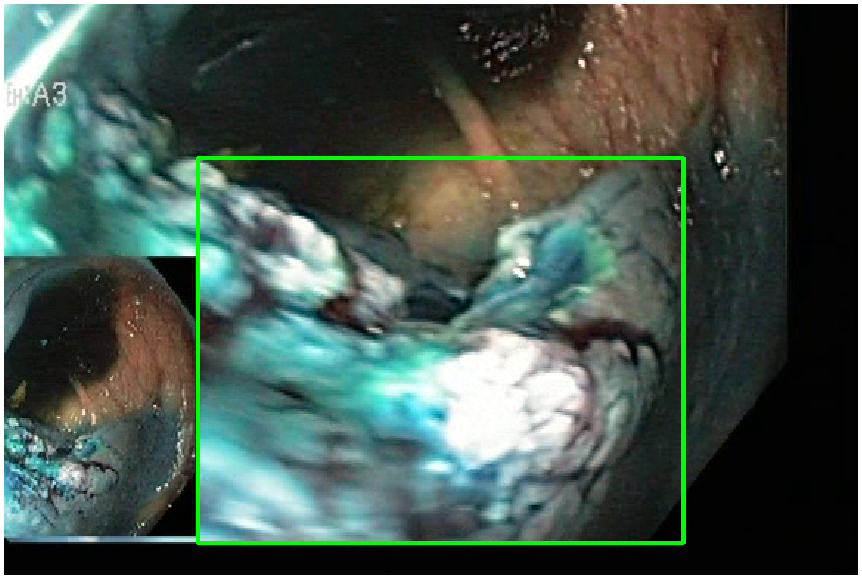
POLYP #8 — LIFTED_POLYP [LOW_RISK] (Tracked across 24 frames)

Feature	Value	Interpretation
Frame Index	1014	Frame 1014 of 2974
Detection Method	Detector Consensus (MedSAM2 Unavailable)	YOLO + RT-DETR agreement (MedSAM2 unavailable for this video)
POLYP TYPE	**LIFTED_POLYP**	Type confidence: 81.5%
Redness Score	0.000	Color-based vascularization
Relative Radius	4.3%	Polyp size as % of frame diagonal
Texture Score	0.522	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	55.7%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #9 — LIFTED_POLYP [LOW_RISK] (Tracked across 22 frames)



Feature	Value	Interpretation
Frame Index	2904	Frame 2904 of 2974
Detection Method	Detector Consensus (YOLO + RT-DETR agreement) (MedSAM2 unavailable for this video)	(MedSAM2 Unavailable)
POLYP TYPE	**LIFTED_POLYP**	Type confidence: 59.4%
Redness Score	0.010	Color-based vascularization
Relative Radius	22.5%	Polyp size as % of frame diagonal
Texture Score	0.134	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	66.8%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #10 — LIFTED_POLYP [LOW_RISK] (Tracked across 22 frames)

Feature	Value	Interpretation
Frame Index	287	Frame 287 of 2974
Detection Method	Detector Consensus (YOLO + RT-DETR agreement (MedSAM2 unavailable for this video)	(MedSAM2 Unavailable)
POLYP TYPE	**LIFTED_POLYP**	Type confidence: 72.7%
Redness Score	0.000	Color-based vascularization
Relative Radius	6.5%	Polyp size as % of frame diagonal
Texture Score	0.460	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	66.0%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #11 — LIFTED_POLYP [LOW_RISK] (Tracked across 21 frames)

Feature	Value	Interpretation
Frame Index	1201	Frame 1201 of 2974
Detection Method	Detector Consensus (YOLO + RT-DETR agreement) (MedSAM2 unavailable for this video)	MedSAM2 Unavailable
POLYP TYPE	**LIFTED_POLYP**	Type confidence: 81.4%
Redness Score	0.000	Color-based vascularization
Relative Radius	10.8%	Polyp size as % of frame diagonal
Texture Score	0.355	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified

Detection Confidence	63.1%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #12 — LIFTED_POLYP [LOW_RISK] (Tracked across 21 frames)

Feature	Value	Interpretation
Frame Index	1502	Frame 1502 of 2974
Detection Method	Detector Consensus (MedSAM2 Unavailable)	YOLO + RT-DETR agreement (MedSAM2 unavailable for this video)
POLYP TYPE	**LIFTED_POLYP**	Type confidence: 84.1%
Redness Score	0.001	Color-based vascularization
Relative Radius	8.4%	Polyp size as % of frame diagonal
Texture Score	0.527	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	58.0%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #13 — LIFTED_POLYP [LOW_RISK] (Tracked across 21 frames)

Feature	Value	Interpretation
Frame Index	1339	Frame 1339 of 2974
Detection Method	Detector Consensus (YOLO + RT-DETR agreement (MedSAM2 unavailable for this video)	MedSAM2 Unavailable
POLYP TYPE	**LIFTED_POLYP**	Type confidence: 78.1%
Redness Score	0.000	Color-based vascularization
Relative Radius	12.1%	Polyp size as % of frame diagonal
Texture Score	0.500	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	49.6%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #14 — LIFTED_POLYP [LOW_RISK] (Tracked across 20 frames)

Feature	Value	Interpretation
Frame Index	623	Frame 623 of 2974
Detection Method	Detector Consensus (YOLO + RT-DETR agreement)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	**LIFTED_POLYP**	Type confidence: 67.2%
Redness Score	0.004	Color-based vascularization
Relative Radius	17.2%	Polyp size as % of frame diagonal
Texture Score	0.158	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	67.1%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #15 — LIFTED_POLYP [LOW_RISK] (Tracked across 20 frames)

Feature	Value	Interpretation
Frame Index	982	Frame 982 of 2974
Detection Method	Detector Consensus (YOLO + RT-DETR agreement) (MedSAM2 unavailable for this video)	MedSAM2 Unavailable
POLYP TYPE	**LIFTED_POLYP**	Type confidence: 81.3%
Redness Score	0.000	Color-based vascularization
Relative Radius	5.2%	Polyp size as % of frame diagonal
Texture Score	0.472	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	63.9%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

CLINICAL IMPRESSION

Summary:
A total of 15 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

Risk Distribution:
• HIGH RISK polyps: 0 (0.0%)
• MEDIUM RISK polyps: 0 (0.0%)
• LOW RISK polyps: 0 (0.0%)

Recommendation:
All detected polyps are classified as LOW RISK. Standard surveillance protocol is recommended.
Technical Details:
• Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM2)
• Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
• Classification: Cluster-specific Decision Tree experts
• Confidence: Based on deep learning + medical heuristics

AI-Based Polyp Detection System
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